

Seattle Weather Project

Does it rain more in Seattle or New York City (NYC)?

1. Introduction

Seattle has a reputation as the rainy city of the United States. But does it really get more rain than NYC? “More rain” can mean how much rain falls or how often it rains. This report compares the two cities on both measures using five years (2018-2022) of daily weather records.

2. The data

The data are daily precipitation measurements from the U.S. National Oceanic and Atmospheric Administration (NOAA), from one weather station in each city (Central Park in NYC, and one station in Seattle). Precipitation includes rain and the water content of melted snow. Each city covers 1,826 days, from January 1, 2018, to December 31, 2022. NYC’s records are complete, but Seattle is missing 190 days (about 10%), so each missing day was filled with the average precipitation for that same calendar day in the other years.

3. Analysis methods

First, I combined the two cities into one table. Next, I compared them in three ways: the total precipitation per year, the average amount per day by month, and the percentage of days with rain by month (a rainy day is any day with measurable precipitation). Looking by month matters because the seasons differ.

Because five years is a small sample, I then ran a statistical test for each month to check whether the difference between the cities is bigger than chance alone could explain. A difference is called significant if there is less than a 5% chance of seeing one that large by luck alone. Stars on the graphs mark those months.

4. Results

- ***Total amount: NYC gets more***

Over the five years, NYC received 270 inches of precipitation and Seattle 207 inches. NYC had more in every year, although the gap was small in 2020 (2.2 inches) and 2022 (3.0 inches).

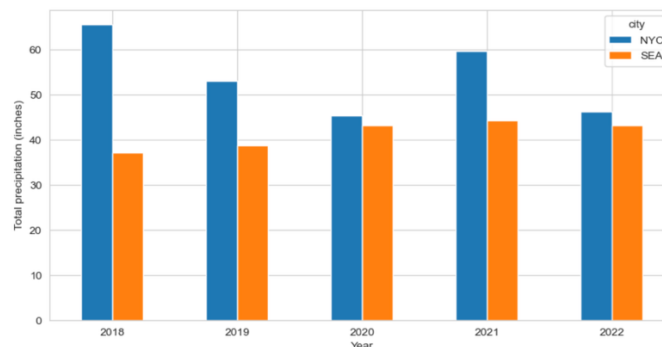


Figure 1. Total precipitation per year

- **The seasons differ**

Looking at the seasons, the picture changes. Seattle gets more rain per day from November to February, while NYC gets more from March to October. The summer gap is large: in July, NYC averages 0.23 inches per day and Seattle only about 0.01.

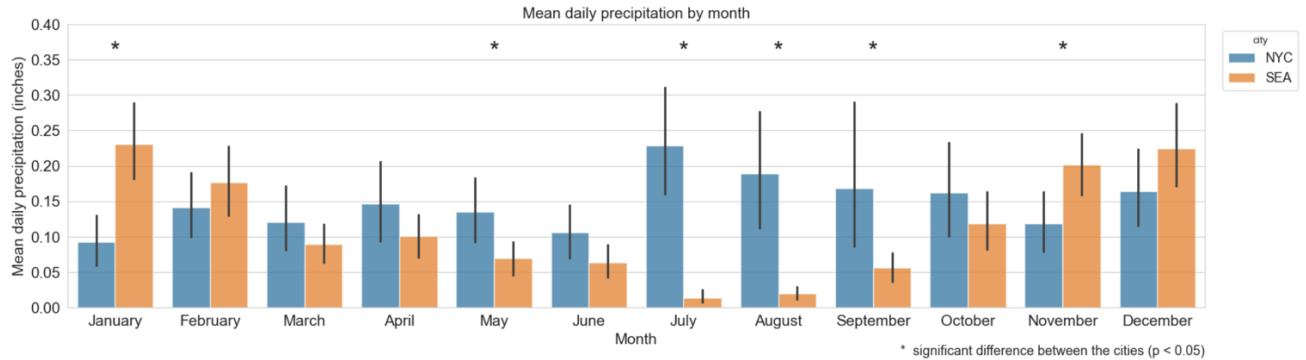


Figure 2. Average precipitation per day, by month

The black lines show how uncertain each average is. A star marks a month with a significant difference between the cities, meaning it is unlikely to be due to chance.

- **Frequency: Seattle rains more often**

Counting rainy days gives a different answer. Seattle has precipitation on about 55% of days, compared with 38% in NYC. From November to February the gap is large: 73%-83% of days in Seattle versus 31-44% in NYC. NYC has more rainy days only in July (41% vs 24%) and August (35% vs 29%), and only July's difference is significant.

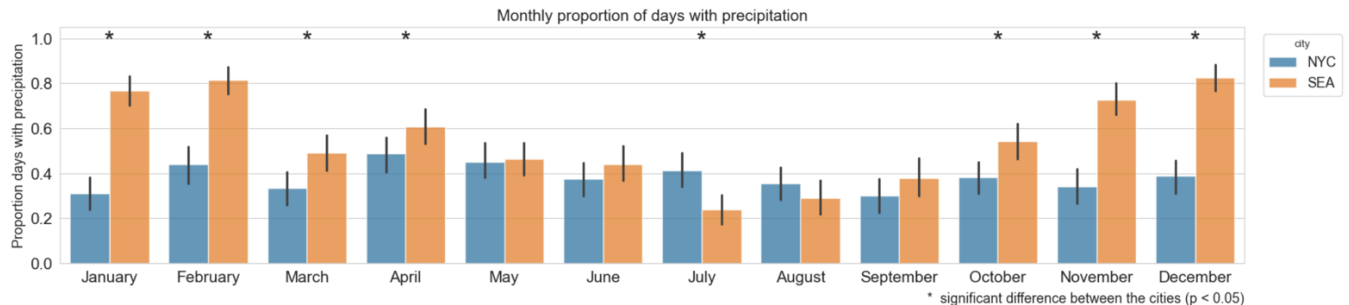


Figure 3. Percentage of days with precipitation, by month. Stars mark months with a significant difference between the cities

5. Conclusion

In conclusion, Seattle rains more often, especially in winter. NYC gets more total precipitation, because its summers are much wetter. Seattle's reputation is therefore fair when it comes to how often it is wet, but not when it comes to how much rain falls.

However, these conclusions have some limits. The analysis covers only five years and one station per city, so it may not represent each whole city. In addition, the 190 filled-in Seattle days count as rainy days, which can make Seattle look slightly rainier (55% of days with them, 51% using only recorded days). The March and April differences in rainy days depend on those filled-in days.